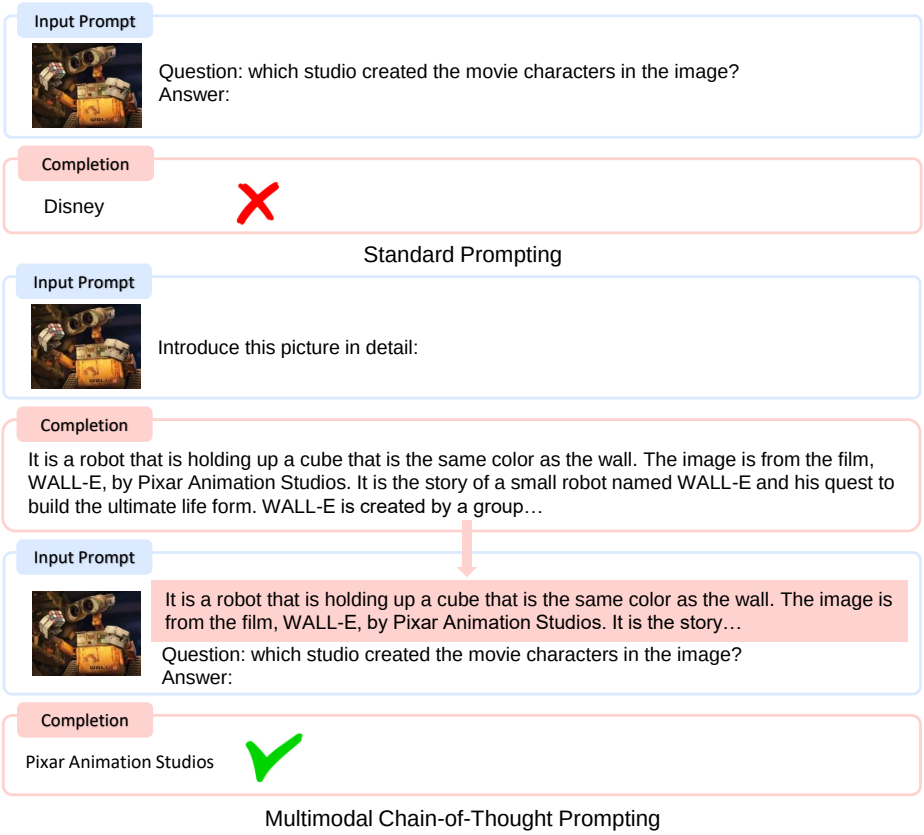

Figure 5: Multimodal Chain-of-Thought prompting enables KOSMOS-1 to generate a rationale first, then to tackle complex question-answering and reasoning tasks.

improve the performance in complex tasks. Motivated by chain-of-thought prompting, we investigate a multimodal chain-of-thought prompting using KOSMOS-1. As illustrated in Figure 5, we break down perception-language tasks into two steps. In the first stage, given an image, we use a prompt to guide the model to generate a rationale. The model is then fed the rationale and a task-aware prompt to produce the final results.

4.5.1 Evaluation Setup

We evaluate the ability of multimodal chain-of-thought prompting on the Rendered SST-2. We use the prompt *"Introduce this picture in detail:"* to generate the content in the picture as the rationale. Then, we use the prompt *"{rationale} Question: what is the sentiment of the opinion? Answer: {answer}"* to predict the sentiment, where the answer is either positive or negative.

4.5.2 Results

We conduct experiments to evaluate the performance of the multimodal chain-of-thought prompting. Table 9 shows that multimodal chain-of-thought prompting achieves a score of 72.9, which is 5.8 points higher than the standard prompting. By generating intermediate content, the model can recognize the text in the images and infer the sentiment of the sentences more correctly.

4.6 Zero-Shot Image Classification

We report the zero-shot image classification performance on ImageNet [DDS+09]. Image classification comprehends an entire image as a whole and aims to assign a label to the image. We map each label to its category name in natural language. The model is prompted to predict the category name to perform zero-shot image classification.

Models	Accuracy
CLIP ViT-B/32	59.6
CLIP ViT-B/16	59.8
CLIP ViT-L/14	64.0
KOSMOS-1	67.1
w/ multimodal CoT prompting	**72.9**

Table 9: Multimodal chain-of-thought (CoT) prompting on Rendered SST-2 task.

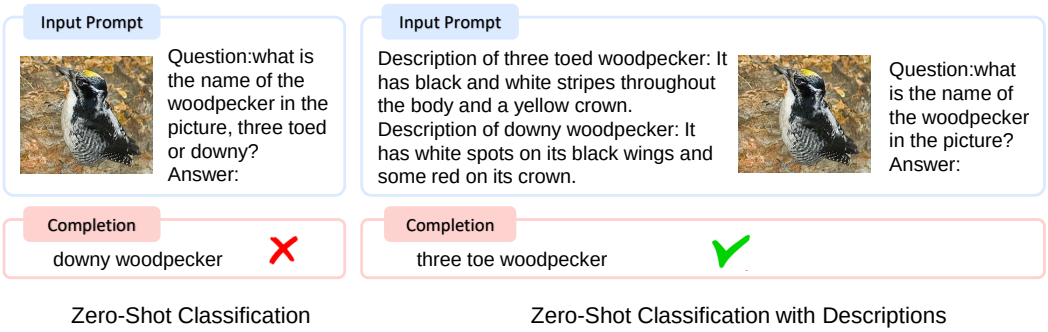

Figure 6: In-context verbal descriptions can help KOSMOS-1 recognize visual categories better.

4.6.1 Evaluation Setup

Given an input image, we concatenate the image with the prompt *"The photo of the"*. The input is then fed into the model to obtain the category name of the image. We evaluate the model on ImageNet [DDS+09], which contains 1.28M training images and 50k validation images in 1k object categories. The prediction is classified as correct if it is exactly the same as the ground-truth category name. The image resolution used for evaluation is 224×224. We use beam search to generate the category names and the beam size is 2.

4.6.2 Results

As shown in Table 10, we report zero-shot results in both constrained and unconstrained settings. The difference between the two settings is whether we use the 1k object category names to constrain the decoding. KOSMOS-1 significantly outperforms GIT [WYH+22] by 4.6% under the constrained setting and 2.1% under the unconstrained setting.

Model	Without Constraints	With Constraints
GIT [WYH+22]	1.9	33.5
KOSMOS-1	**4.0**	**38.1**

Table 10: Zero-shot image classification on ImageNet. For the results with constraints, we use the 1k ImageNet object category names for constrained decoding. We report top-1 accuracy scores.

4.7 Zero-Shot Image Classification with Descriptions

The standard approach of image classification as above is to prompt the model for the specific name of the object depicted in the image. However, there are also some classification rules customized for different users and scenarios, such as the refined classification of complex animal subspecies. We can utilize natural language descriptions to guide KOSMOS-1 to distinguish images in the zero-shot setting, which makes the decision process more interpretable.

Category 1		Category 2	
three toed woodpecker		downy woodpecker	
	It has black and white stripes throughout the body and a yellow crown.		It has white spots on its black wings and some red on its crown.
Gentoo penguin		royal penguin	
	It has a black head and white patch above its eyes.		It has a white face and a yellow crown.
black throated sparrow		fox sparrow	
	It has white underparts and a distinctive black bib on the throat.		It has a reddish-brown plumage and a streaked breast.

Table 11: The detailed descriptions of different categories for in-context image classification.

4.7.1 Evaluation Setup

Following CUB [WBW+11], we construct a bird classification dataset that contains images and natural-language descriptions of categories. The dataset has three groups of binary image classification. Each group contains two animal categories with similar appearances. Our goal is to classify images given the categories' descriptions. Table 11 presents the data samples. The first group is from [WBW+11], while the other two groups are collected from the website. Each category contains twenty images.

The evaluation procedure is illustrated in Figure 6. For the zero-shot setting, we provide detailed descriptions of two specific categories and use the template *"Question:what is the name of {general category} in the picture? Answer:"* to prompt the model for the specific category name in an open-ended manner. To evaluate the effect of providing verbal descriptions in context, we also implement a zero-shot baseline without prompting descriptions. Instead, we provide the corresponding specific names in the prompt.

4.7.2 Results

The evaluation results are shown in Table 12. We observe that providing descriptions in context can significantly improve the accuracy of image classification. The consistent improvements indicate that KOSMOS-1 can perceive the intentions of instructions and well align the concepts in language modality with visual features in vision modality.

Settings	Accuracy
Without Descriptions	61.7
With Descriptions	**90.0**

Table 12: Results of zero-shot image classification without and with verbal descriptions.

4.8 Language Tasks

The models are evaluated on the language tasks given task instructions (i.e., zero-shot) or several demonstration examples (i.e., few-shot). Text inputs are directly fed into the models as in vanilla language models.